



Impact of Educational Programs and VR Technologies on the Environmental Behavior of the Population in Areas with High Pollution Levels: Case Study of Kazakhstan

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Abstract: The study goal was to evaluate the effectiveness of school eco-clubs and digital technologies as instruments for fostering environmental responsibility amid the environmental challenges in rural areas of Southeast Kazakhstan. A descriptive-analytical design with mixed methods was applied. The sample consisted of 186 “student–parent” pairs from 10 schools in the Almaty Region and 30 experts. Data was collected from September 2023 to April 2024 using validated questionnaires, focus groups, and expert interviews. A statistically significant relationship was identified between the duration of participation in eco-clubs and the level of students’ environmental knowledge ($p < 0.01$) and between the intensity of intergenerational communication and changes in family practices: safe handling of agrochemicals (62.2% of families), composting organic waste (55.1%), and the use of organic fertilizers (38.8%). Regression analysis showed that practical demonstrations ($\beta = 0.42$, Cohen’s $d = 0.89$) and virtual reality (VR) technologies ($\beta = 0.35$, Cohen’s $d = 0.74$) are the strongest predictors of changes in environmental behavior ($R^2 = 0.51$, $p < 0.001$). VR outperforms traditional methods: knowledge acquisition is 35% higher (4.6 versus 3.4 points), and practical application is twice as high (89.3% and 45.2%, respectively). The study confirms the effectiveness of school eco-clubs as a catalyst for environmental changes in rural families. The authors demonstrate the potential of virtual technologies in compensating for the limitations of environmental education in remote agricultural regions.

Keywords: Eco-cubs; Intergenerational knowledge transfer; Digital technologies; Environmental education; Environment

1 Introduction

Modern global challenges necessitate a major revision of approaches to fostering environmental responsibility [1], especially in the context of developing educational programs, the intergenerational transmission of knowledge and values, and digital technologies [2]. Researchers increasingly point out the need to break the vicious cycle in environmental education. Traditionally, children adopt environmental habits from their parents, mimicking their attitudes toward nature and their ways of interacting with the environment [3, 4]. However, if parents demonstrate environmentally destructive behavior, such as careless waste disposal, wasteful resource use, or ignoring environmental issues, children adopt these same unsustainable practices. Thus, the intergenerational transmission of environmentally harmful habits hinders the development of environmental responsibility [5]. Traditional educational approaches, relying on formal teaching methods (teacher lectures, textbook study, memorization of facts) and focusing exclusively on the one-way transfer of knowledge from the older generation to the younger, prove insufficient for building a qualitatively new ecological mindset necessary to address global environmental issues and achieve the principles of sustainable development [6, 7].

In search of effective mechanisms to change the environmental consciousness of the younger generation, researchers turn their attention to educational programs and initiatives capable of creating conditions for the mutual intergenerational exchange of knowledge and experience. This involves not only adults teaching children but also children becoming a source of new environmental knowledge for their parents. Of note are programs in which children

initiate environmental changes in families, actively promoting new environmental knowledge and practices among parents [3, 7, 8]. This approach becomes particularly valuable in the framework of school environmental education, which creates the conditions for a systematic impact on the development of environmental culture, especially when it utilizes modern digital technologies [9, 10].

One of the most promising tools for such an approach is school eco-clubs, which are voluntary student associations raising environmental awareness and promoting conservation activities through practical participation [11]. Eco-clubs motivate their participants, particularly school students, to go beyond the school environment, involving families and local communities in environmental initiatives.

2 Literature Review

2.1 International Organizational Models of Eco-Clubs

Eco-clubs are growing in popularity all over the world. Furthermore, international exchange programs are becoming an important component in the work of eco-clubs. The Foundation for Environmental Education [12], uniting over 100 organizations in 83 countries, shows success in global educational initiatives, such as the Eco-Schools program. In the program, participants study environmental practices from different countries and exchange experiences with their peers.

The analysis of international experience reveals a great variety of organizational models of eco-clubs, differing in goals, methods of operation, and the degree of integration into the educational process. Ramadoss and Poyyamoli [11] identify three main models of eco-clubs based on their functional role.

The educational and informational model focuses on increasing environmental literacy by studying environmental issues, discussing specialized literature, and conducting lectures [13]. Participants in such clubs examine environmental cases, analyze scientific publications, and prepare presentations on current environmental issues. The practice-oriented model emphasizes direct participation in environmental protection activities: organizing ecological events, monitoring the state of the environment [14], and creating school ecosystems (gardens, greenhouses, and compost systems). Alexandar and Poyyamoli [15] note that such clubs demonstrate higher effectiveness in fostering sustainable environmental practices.

The social activism model aims at mobilizing the school and local community to address environmental issues [16]. Participants organize awareness campaigns, interact with local authorities and environmental organizations, and initiate changes in school and family practices.

2.2 Educational Methods Used in Eco-Clubs

Modern eco-clubs use a wide range of educational methods, from traditional [17] to innovative technological solutions [18, 19].

Traditional methods include discussions of environmental literature, debates on environmental topics, and themed quizzes and competitions. Kerret et al. [20] stress the importance of creating a positive emotional connection with environmental issues through group discussions and collaborative problem-solving [21].

Game-based and simulation methods are gaining popularity due to their ability to engage students in active learning. Role-playing games allow participants to model various environmental scenarios and make decisions on behalf of different stakeholders (industrialists, ecologists, residents, and government officials). This approach promotes the development of systems thinking and an understanding of the complexity of environmental issues.

Digital and virtual technologies open new opportunities for environmental education [22]. For instance, virtual reality (VR) technologies allow students to “visit” inaccessible ecosystems, observe the consequences of environmental disasters, and model future climate change scenarios [23]. Ardoin et al. [24] in their meta-analysis found that technologically enhanced programs show higher student engagement and better material retention.

The successful integration of eco-clubs and school education requires support from school administration, the preparedness of teaching staff, material and technical support, and alignment with the curriculum [25, 26]. Moreover, eco-clubs face institutional, resource, and motivational barriers [27].

2.3 Eco-Club Development Practices in Central Asia

Eco-clubs are developed particularly actively in regions with relatively low living standards and an underdeveloped system of environmental education, including Asian countries [15, 28]. In small towns and rural areas of these regions, environmental problems often escalate faster than the mechanisms for their resolution are developed. This makes school eco-clubs an important tool not only for education but also for social mobilization to address local environmental problems [29].

In the context of Kazakhstan, eco-clubs are considered an important component of the national project for greening education [30]. Southeast Kazakhstan is a prime example of a region facing serious environmental challenges that require innovative approaches in environmental education. The region is characterized by critical air pollution and a growing water shortage, which significantly impact public health and socio-economic development [31–33].

Although there are many international and regional programs, earlier studies did not examine how eco-clubs work in pollution-affected rural communities, nor did they analyze the pathways through which children influence their parents' environmental behavior. Furthermore, prior studies rarely integrated local ecological problems into program evaluation. Thus, despite numerous attempts to develop the topic of the possibilities of eco-clubs, there is a noticeable lack of research on eco-clubs in developing regions with acute environmental problems, such as Central Asia. The present study aims to fill these gaps through a comprehensive analysis of the role of eco-clubs in fostering environmental responsibility through intergenerational knowledge transfer with the involvement of digital technologies in the context of Southeast Kazakhstan.

2.4 Research Goal and Objectives

Research goal: To determine the impact of educational programs and digital technologies on the environmental behavior of the population in areas with high levels of pollution in the context of Southeast Kazakhstan.

Research objectives:

- To determine the impact of participation in eco-clubs on the environmental knowledge and behavior of students and their parents.
- To identify the most effective mechanisms of intergenerational knowledge transfer in the context of eco-club activities utilizing digital technologies.
- To evaluate the potential of integrating local environmental issues into eco-club programs to enhance their effectiveness.

Research hypothesis: School eco-clubs that integrate localized environmental content through practical experience and digital instruments and systematically engage participants' families in club activities show significantly greater effectiveness in fostering sustainable pro-environmental practices among students and parents compared to the traditional methods of environmental education.

3 Methods

3.1 Design and Background

The study employed a descriptive-analytical design using mixed methods to examine the role of intergenerational transfer of environmental knowledge. Southeast Kazakhstan was chosen due to the representativeness of local environmental problems, the presence of schools meeting the research goal, and accessibility for the study. The focus on schools in Southeast Kazakhstan allowed us to examine the impact of eco-clubs in the context of the agricultural environmental issues prevalent in the region. Data collection was conducted from September 2023 to April 2024.

Southeast Kazakhstan was chosen for the study due to the fact that educational cases show a rise in the implementation of programs focused on waste recycling. These programs include the study of recycling technologies, the organization of separate waste collection in schools, and creating items from recycled materials. Yelubayeva et al. [28] highlight the success of the Green School project in Almaty, where the incorporation of waste sorting contributed to the environmental responsibility of students. However, in urban and rural areas, the concentrations of major pollutants significantly exceed WHO recommendations [31, 34], and air pollution increases the prevalence of respiratory and cardiovascular diseases [32]. For these reasons, the state has developed a dedicated educational program to create eco-clubs focused on studying the connection between air quality, water, natural areas [35], and public health [36] to engage students and their families in jointly addressing these issues.

3.2 Sample

Out of the 25 schools were purposefully selected based on the criteria of the eco-club functioning for at least 2 years, geographical diversity, and the school's readiness to cooperate. From each school, 18–20 students participating in the eco-club were selected by random sampling (ages 16–18, with a minimum participation period of 6 months). The final sample consisted of 186 student-parent pairs (93% response rate). One parent from a family was invited to participate, preferably the one who was more involved in environmental practices. If both parents were present, participation was voluntary.

The expert assessment involved 30 specialists: coordinators of eco-clubs ($n = 10$), representatives of environmental NGOs ($n = 10$), and education management specialists ($n = 10$).

3.3 Research Methods

3.3.1 Quantitative methods

Structured questionnaires for students (Appendix 1) and their parents (Appendix 2) comprising the blocks of environmental knowledge, activities in the eco-club, environmental behavior, and intergenerational communication ("Communication with parents on environmental topics" in the student questionnaire).

Expert survey—rating 10 factors determining the effectiveness of eco-clubs on a 5-point scale (Appendix 3).

3.3.2 Qualitative methods

Focus groups: four focus groups with students and four with parents (8–10 people per group) lasting 60–90 minutes (Appendix 4).

The validity and reliability of the questionnaires were verified through a pilot test. This involved 30 students from three schools, who were not part of the main sample. By the results of the pilot study, revision was made to better formulate questions. The internal consistency of the questionnaire scales was assessed using Cronbach's α : $\alpha = 0.82$ for the environmental knowledge scale and $\alpha = 0.79$ for the environmental behavior scale. For the intergenerational communication scale, $\alpha = 0.76$. These results confirm that the instruments exhibited acceptable reliability.

3.4 Data Analysis

The quantitative data were analyzed using descriptive statistics, cross-tabulation, and Spearman's correlation analysis. The consistency of expert assessments was tested via Kendall's coefficient of concordance. Cross-group differences were analyzed with Pearson's χ^2 test (for categorical variables) and Student's t-test (for continuous variables). The qualitative data was subjected to content analysis, which included identifying key themes and compiling a pivot table with participants listed in rows and the discussed topics in columns to systematize the collected qualitative data.

To assess the practical significance of the results, effect size indicators were used: Cohen's d for comparing mean values between groups and Cramer's V for analyzing associations between categorical variables. Multiple linear regression was applied to identify the predictors of changes in families' environmental behavior, where the dependent variable was the change in family environmental practices (measured using a composite 10-item scale). The independent variables included the duration of participation in the eco-club, the frequency of intergenerational communication, the degree of integration of local content, and the use of VR.

3.5 Ethical Aspects and Limitations

Informed consent was obtained from all participants in the study, and the data was made anonymous. The key limitations of the study limiting the extrapolation of findings include the lack of a control group, the subjectivity of self-assessments, and the specificity to the region (only the Almaty Region).

4 Results

The results are presented according to the set objectives and are centered around testing the hypothesis about the role of eco-clubs in developing environmental responsibility through the mechanisms of intergenerational knowledge transfer.

4.1 Influence of Participation in Eco-Clubs on Participants' Environmental Knowledge

The relationship between students' environmental knowledge of key local environmental problems and the duration of their participation in eco-clubs is analyzed in Table 1.

The results demonstrate a statistically significant direct relationship between the duration of participation in eco-clubs and the level of environmental knowledge across all studied categories, confirming the effectiveness of long-term participation in the programs.

Table 1. Level of students' environmental knowledge depending on the period of participation in eco-clubs ($n = 186$)

Category of Knowledge	6–12 Months ($n = 72$)	1–2 Years ($n = 77$)	More than 2 Years ($n = 37$)	<i>p</i> -Value
Air pollution sources				
High level	54.2% (39)	70.1% (54)	86.5% (32)	<0.01
Average level	31.9% (23)	23.4% (18)	13.5% (5)	
Low level	13.9% (10)	6.5% (5)	0% (0)	
Water conservation methods				
High level	61.1% (44)	75.3% (58)	89.2% (33)	<0.01
Average level	27.8% (20)	20.8% (16)	8.1% (3)	
Low level	11.1% (8)	3.9% (3)	2.7% (1)	
Waste recycling technologies				
High level	38.9% (28)	54.5% (42)	70.3% (26)	<0.01
Average level	38.9% (28)	36.4% (28)	27.0% (10)	
Low level	22.2% (16)	9.1% (7)	2.7% (1)	

Note: The statistical significance of differences was assessed using Pearson's χ^2 test. High level: $\geq 80\%$ of correct answers; average level: 50%–79%; low level: $< 50\%$.

A comparative analysis of changes in the environmental behavior of the families of eco-club participants depending on the intensity of intergenerational communication is presented in Table 2.

Table 2. Changes in the environmental behavior of families depending on the frequency of discussing environmental topics ($n = 186$)

Environmental Practice	High Frequency (<i>n</i> = 98)	Low Frequency (<i>n</i> = 88)	<i>p</i> -Value
Safe storage and disposal of agricultural chemicals			
Significant changes	62.2% (61)	33.0% (29)	<0.001
Moderate changes	29.6% (29)	35.2% (31)	
Insignificant or no changes	8.2% (8)	31.8% (28)	
Composting organic waste instead of incineration			
Significant changes	55.1% (54)	27.3% (24)	<0.001
Moderate changes	35.7% (35)	43.2% (38)	
Insignificant or no changes	9.2% (9)	29.5% (26)	
Using homemade organic fertilizers			
Significant changes	38.8% (38)	15.9% (14)	<0.01
Moderate changes	39.8% (39)	31.8% (28)	
Insignificant or no changes	21.4% (21)	52.3% (46)	
Participation in local environmental campaigns to preserve water resources			
Significant changes	25.5% (25)	11.4% (10)	<0.05
Moderate changes	42.9% (42)	26.1% (23)	
Insignificant or no changes	31.6% (31)	62.5% (55)	

Note: High frequency of communication—daily or 2–3 times a week; low frequency—once a week, once a month, or no discussions. Statistical significance was determined using Pearson's χ^2 test.

The data reveal a statistically significant relationship between the intensity of intergenerational communication on environmental issues and changes in family environmental practices, confirming the importance of systematic family engagement.

Table 3 presents the analysis of the impact of integrating local environmental content (specific environmental issues faced by the residents of the locality where the eco-club is situated) on the effectiveness of eco-club programs.

Table 3. Comparative effectiveness of eco-clubs with high and low integration of local content ($n = 186$)

Effectiveness Indicator	High Integration (<i>n</i> = 106)	Low Integration (<i>n</i> = 80)	<i>p</i> -Value
Students' environmental knowledge (mean score)			
Local problems	4.2 ± 0.8	3.1 ± 0.9	<0.001
Global problems	3.8 ± 0.7	3.7 ± 0.8	0.243
Behavioral changes in families (%)			
Significant changes	67.9% (72)	35.0% (28)	<0.001
Moderate changes	26.4% (28)	46.3% (37)	
Insignificant or no changes	5.7% (6)	18.7% (15)	
Practical application of knowledge (%)			
Regular use	73.6% (78)	45.0% (36)	<0.001
Occasional use	22.6% (24)	40.0% (32)	
Rare or no use	3.8% (4)	15.0% (12)	
Motivation to participate (mean score)	4.5 ± 0.6	3.9 ± 0.8	<0.001

Note: High integration: systematic inclusion of local environmental issues in 70% or more of the classes; low integration: less than 30% of the classes. Knowledge was assessed on a 5-point scale. Statistical significance was assessed using Student's *t*-test for continuous variables and χ^2 test for categorical variables.

The results demonstrate a statistically significant advantage of eco-clubs with high integration of local content across all key effectiveness indicators, except for knowledge of global environmental issues.

The correlation analysis of relationships between the different mechanisms of intergenerational knowledge transfer and the effectiveness of the program is presented in Table 4.

Table 4. Correlation analysis of intergenerational knowledge transfer mechanisms ($n = 186$)

Transfer Mechanism	Changes in Parents' Knowledge	Changes in Family Behavior	Consistency of Changes
Frequency of discussions on local (community) issues in various formats	0.54***	0.62***	0.48***
Practical experience in applying the obtained knowledge in daily life	0.67***	0.71***	0.58***
Motivation for behavior change after collaborative learning using virtual technologies	0.63***	0.68***	0.61***
Joint participation in eco-initiatives	0.45***	0.56***	0.52***
Transfer of specific water and energy conservation techniques	0.59***	0.65***	0.55***
Emotional engagement	0.41***	0.47***	0.44***
Exchange of experience with other families as part of the eco-club	0.33**	0.29**	0.25*

Note: Spearman's correlation coefficients. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. The consistency of changes was assessed 6 months after the beginning of the study.

The most effective mechanisms of intergenerational knowledge transfer are practical home demonstrations and the transmission of specific skills, which show the strongest correlations with all effectiveness indicators.

The results of expert analysis of the factors determining the effectiveness of intergenerational knowledge transfer in eco-clubs are summarized in Table 5.

Table 5. Expert assessment of factors in the effectiveness of intergenerational knowledge transfer ($n = 30$)

Factor	Mean	SD	Rank
Practical applicability of knowledge	4.87	0.35	1
Systematicity of household discussions	4.73	0.45	2
Collaborative learning using virtual technologies	4.68	0.42	3
Connection with everyday family problems	4.67	0.48	4
Active engagement of parents	4.60	0.50	5
Illustrative and specific examples	4.53	0.51	6
Emotional engagement of children	4.47	0.57	7
Support of family initiatives by the school	4.33	0.61	8
Using local context	4.27	0.64	9
Inter-family interaction	3.93	0.70	10

Note: Assessment on a 5-point scale (1 = completely unimportant, 5 = extremely important). Kendall's coefficient of concordance $W = 0.79$, $p < 0.001$.

The high consistency of expert opinions confirms the reliability of the results, which highlights the practical applicability of knowledge as a key factor in the successful intergenerational transfer of environmental knowledge and skills.

Table 6 presents a comparison of the effectiveness of different educational methods used in eco-clubs.

Table 6. Comparative effectiveness of educational methods in eco-clubs ($n = 186$)

Educational Method	Number of Participants	Knowledge Assimilation (Mean Score)	Practical Application (%)	Motivation to Participate (Mean Score)
VR technologies and virtual excursions	42	4.6 $\pm$ 0.4***	89.3% (37)***	4.8 $\pm$ 0.3***
Role-playing games and simulations	78	4.2 $\pm$ 0.6**	71.8% (56)**	4.4 $\pm$ 0.5**
Practical demonstrations and experiments	96	4.0 $\pm$ 0.7*	68.8% (66)*	4.2 $\pm$ 0.6*
Traditional lectures and discussions	124	3.4 $\pm$ 0.8	45.2% (56)	3.6 $\pm$ 0.7
Digital monitoring apps	35	4.1 $\pm$ 0.5**	74.3% (26)**	4.3 $\pm$ 0.4**
International online exchange	28	4.3 $\pm$ 0.6**	78.6% (22)**	4.5 $\pm$ 0.5**

Note: VR, virtual reality. Participants could use several methods simultaneously. Statistical significance of differences with the control group (traditional methods): * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ (Student's t-test for continuous variables, χ^2 test for categorical variables). Knowledge assimilation was assessed on a 5-point scale through testing. Practical application refers to the share of participants demonstrating sustainable changes in behavior after 3 months.

Thus, the conducted analysis demonstrates the superiority of interactive teaching methods over traditional lecture-based approaches.

4.2 Regression Analysis of Predictors of Environmental Behavior

Multiple linear regression revealed a statistically significant model predicting changes in families' environmental behavior ($F(4,181) = 47.3, p < 0.001, R^2 = 0.51$). The model explains 51% of the variance of the dependent variable, which indicates a significant contribution of the studied predictors. The strongest predictors were the frequency of practical demonstrations of environmental skills at home ($\beta = 0.42, p < 0.001$, Cohen's $d = 0.89$) and the use of VR in the educational process ($\beta = 0.35, p < 0.001$, Cohen's $d = 0.74$). Duration of participation in the eco-club ($\beta = 0.21, p < 0.01$) and integration of local content ($\beta = 0.18, p < 0.05$) showed a less pronounced effect, although statistically significant. The effect sizes for the main predictors fall within the "large" range according to Cohen's classification, confirming the practical significance of the identified relationships.

5 Discussion

The results confirm the main hypothesis of the study regarding the high effectiveness of the educational program for developing school eco-clubs, which integrates localized environmental content through practical experience and digital tools. The observed direct correlation between the duration of participation in eco-clubs and the level of environmental knowledge ($p < 0.01$ for all categories) shows the cumulative effect of environmental education in rural settings. Importantly, the greatest increase in knowledge was observed with participation for over 2 years, suggesting the need for long-term educational programs.

The results of the regression analysis quantitatively confirm the hierarchy of effectiveness factors. The model explains 51% of the variability in changes in families' environmental behavior, which is a substantial indicator for research in the field of education. Practical demonstrations of environmental skills show the strongest effect, which aligns with Bandura's social learning theory, emphasizing the importance of behavior modeling. The effect size (Cohen's $d = 0.89$) confirms not only the statistical significance but also the practical relevance of this mechanism of knowledge transfer.

The statistically significant relationship between the intensity of intergenerational communication and changes in family environmental practices ($p < 0.001$ for all studied practices) confirms the role of children as catalysts for environmental changes in rural families. Especially pronounced changes were found in the practices of the safe use of agrochemicals (62.2% of families with a high level of communication vs. 33.0% with a low level) and composting organic waste (55.1% vs. 27.3%), which is critically important for agricultural regions.

The standardized regression coefficient $\beta = 0.35$ for the use of VR confirms its significant independent contribution to the development of environmental behavior, even when controlling other factors. The effect size indicates the significance of the impact, which is particularly important for educational programs.

VR and collaborative virtual learning showed a statistically significant advantage across all indicators: knowledge acquisition (4.6 vs. 3.4 points, $p < 0.001$), practical application (89.3% vs. 45.2%, $p < 0.001$), and motivation (4.8 vs. 3.6 points, $p < 0.001$). The high rank of the factor of "joint learning using virtual technologies" in the expert assessment (3rd place, 4.68 points) confirms the particular significance of technological solutions in rural areas, where access to a variety of environmental objects for direct study is limited.

The established effectiveness of the intergenerational transfer of environmental knowledge in the rural context adds to the results of Lawson et al. [3], which demonstrate the ability of children in urban settings to influence their parents' environmental beliefs. Our results reveal an even more pronounced effect in rural settings: up to 62.2% of families demonstrated significant changes in critically important agricultural practices compared to the 35%–45% reported in international literature for urban families. This emphasizes that the population of the rural areas in Kazakhstan relies not on modern, scientifically proven practices or legal and environmental norms [37] but rather on established traditions in their environmental behavior.

The implementation of such programs should follow a phased, evidence-based approach. At the initial stage, the school introduces the program in a basic format, relying on available resources and an adapted set of activities. After completing the first cycle, an evaluation is conducted to assess its impact on students' environmental knowledge and the ecological practices of their families. Schools that demonstrate a confirmed positive effect receive priority funding for program expansion. Based on the analysis of achieved results and identified challenges, an individualized development plan is formulated. This plan includes targeted provision of equipment, such as digital tools and VR, as well as methodological and organizational support. This "pilot–evaluation–equipment" model allows resources to be directed toward educational institutions where the program yields the highest returns, ensuring efficient allocation of funding and strengthening the overall effectiveness of environmental education in rural settings.

The high effectiveness indicators of practical experience in applying knowledge in everyday life ($r = 0.71$) and the transfer of specific water and energy conservation skills ($r = 0.65$) confirm the particular importance of applied skills [38] for rural families, which aligns with the special role of specific applicable skills [8]. In a rural context,

such skills have direct economic significance [39] in the development of agricultural resources, which boosts the motivation to acquire them [40].

The greater effectiveness of VR in our study compared to the meta-analysis by Yessenbayeva et al. [41] may be explained by the specifics of rural conditions, where virtual technologies compensate for the limited opportunities to observe various ecosystems [42] and environmental processes directly. To rural school students, VR excursions offer a unique opportunity to “visit” various biomes and study global environmental processes not found in their everyday environment, which may further increase their interest in Science, Technology, Engineering, and Mathematics (STEM) [43, 44].

The specifics of rural ecological practices, such as the safe handling of agrochemicals, composting instead of burning waste, and using organic fertilizers, demonstrate the need to adapt eco-club programs to local conditions, where people neglect environmental and ecological issues in pursuit of their goals [45]. In the future, sustainable environmental behavior can improve not only the local environment but also the safety of products and crops [46]. The participation of families in local environmental campaigns for the conservation of aquatic biological resources showed a statistically significant, albeit the smallest, effect (25.5% versus 11.4%). This result can be attributed to the traditionally high social cohesion of rural communities [47] and the recognition of the significance of this issue for the region. However, the events themselves and their focus, organization, and financial aspects are criticized by the population. This only confirms research results warning that substantial financial support [48] and organization are the primary conditions for achieving the desired effect in solving major environmental problems, followed by increasing the motivation of the population [49].

The four key predictors (practical demonstrations, VR, duration of participation, and local content) explain 51% of the variance in changes in families’ environmental behavior ($R^2 = 0.51$), confirming their important role while recognizing the multifactorial nature of behavioral change. Nevertheless, the remaining 49% of unexplained variance suggests the influence of additional factors, which may include family values, socio-economic status, and support from the local community, indicating the need for further research.

Here, we should note several important limitations of our study. First, the lack of a control group does not allow us to completely rule out the influence of external factors on the observed changes. Second, the exclusive focus on rural areas limits our ability to extrapolate the results to urban conditions and other regions of Kazakhstan and Central Asia. Finally, the generalized presentation of data for the entire region could have obscured local differences between rural settlements.

Of particular interest is a more detailed study of the potential of VR in environmental education in rural schools, including in the development of specialized content for studying agricultural ecosystems, sustainable land use, and a careful attitude toward nature, which is critical for achieving the principles of sustainable development [50].

6 Conclusions

The results confirm the need for a systematic approach to the development of eco-clubs, with an emphasis on long-term participation, active family involvement, and the integration of modern technologies into the educational process.

Regression analysis identified the key predictors of change: practical demonstrations, VR, and joint virtual learning involving children and parents showed the strongest effects compared to traditional methods. This finding is of considerable scientific interest and opens new opportunities for the technological modernization of environmental education. Practical demonstrations and the transfer of specific skills in the home environment demonstrated the highest effectiveness among the mechanisms of intergenerational knowledge transfer. This highlights the importance of reinforcing sustainable practices of responsible environmental behavior among parents.

The scientific significance of the study lies in its contribution to the understanding of the mechanisms of intergenerational transmission of ecological knowledge and the demonstration of the potential of modern technologies in environmental education. Furthermore, the results contribute to the development of the theory of environmental education and provide an empirical basis for further research in this field.

The practical significance of the results lies in the opportunity to apply the identified patterns to improve environmental education programs in developing regions. In addition, the established high effectiveness of integrating local environmental content can be leveraged to adapt international educational programs to the specific conditions of Central Asia.

Author Contributions

Conceptualization, K.B. and A.B.; methodology, K.B.; software, K.B.; validation, K.B. and A.B.; formal analysis, K.B.; investigation, K.B.; resources, A.B.; data curation, K.B.; writing—original draft preparation, K.B.; writing—review and editing, K.B. and A.B.; visualization, K.B.; supervision, A.B.; project administration, A.B. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

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Conflicts of Interest

The authors declare that they have no conflicts of interest.

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Appendix

Appendix 1

Survey of students participating in eco-clubs

Instructions: Dear participant! Please answer the following questions about your participation in the eco-club and environmental knowledge. Your answers will help us improve the work of the eco-club. The survey is completely anonymous.

Block 1. Demographic information

1. Age: _____

2. How long have you been participating in the eco-club's activities?

☐ <6 months ☐ 6–12 months ☐ 1–2 years ☐ >2 years

Block 2. Environmental knowledge

5. What are the main sources of air pollution in Southeast Kazakhstan? (select all that apply)

☐ Transport ☐ Industrial and agricultural facilities ☐ Private sector ☐ Other _____

6. How much does air pollution in Southeast Kazakhstan exceed WHO standards?

☐ By 2 times ☐ By 3–4 times ☐ By 4–5 times ☐ Not sure

7. What percentage of water use in Kazakhstan does agriculture account for?

☐ 20%–30% ☐ 40%–50% ☐ 60%–70% ☐ >70%

8. What water conservation methods are you familiar with?

☐ Turning off the tap when brushing teeth ☐ Taking a bath instead of showering ☐ Using the washing machine only when a full load has accumulated ☐ Collecting rainwater for watering plants ☐ Installing water-saving attachments ☐ Keeping track of the time spent in the shower ☐ Reusing water (for example, after washing vegetables) ☐ Not sure

9. What methods of protection against polluted air are you familiar with? (select all that apply)

☐ Using protective masks ☐ Using air purifiers at home ☐ Proper room ventilation ☐ Monitoring the air quality index ☐ Not sure

Block 3. Activities in the eco-club

10. What technologies have been applied in your eco-club? (select all that apply)

☐ VR glasses for virtual excursions ☐ Mobile apps for environmental monitoring ☐ AI programs ☐ Online platforms for international exchange ☐ Digital microscopes ☐ Air/water quality sensors ☐ None of the above

11. Have you participated in role-playing games on environmental topics?

☐ Often (more than 5 times) ☐ Sometimes (2–4 times) ☐ Rarely (1 time) ☐ Never

12. What practical skills have you learned in the eco-club? (select all that apply)

☐ Sorting and recycling waste ☐ Making things from recycled materials ☐ Plant growing ☐ Improving air/water quality ☐ Energy conservation in daily life ☐ Planning environmental initiatives

13. Have you studied the ways to recycle various materials in the eco-club?

☐ Yes, we studied the technologies in detail ☐ Only the key principles ☐ Only superficially ☐ No, we didn't

14. Have you discussed the methods of personal protection against environmental pollution in the eco-club?

☐ Yes, regularly ☐ Yes, sometimes ☐ Yes, rarely ☐ No

Block 4. Environmental behavior

15. What do you think should change in the behavior of people in your region to improve the environmental situation? _____

16. What efforts does your family make to save water? (select all that apply)

☐ Turning off the tap when brushing teeth ☐ Using the washing machine only when a full load has accumulated ☐ Collecting rainwater ☐ Keeping track of the time spent in the shower ☐ Other

17. Have your habits changed as a result of your involvement in the eco-club? (select all that apply)

☐ I started saving water more ☐ I started sorting waste ☐ I use public transport more ☐ I buy more eco-friendly products ☐ I keep track of air quality ☐ No changes

Block 5. Communication with parents on environmental topics

18. How often do you discuss environmental topics with your parents?

☐ Daily ☐ 2–3 times a week ☐ Once a week ☐ Once a month ☐ Never

19. Do you tell your parents about what you do in the eco-club?

☐ I regularly share the new knowledge ☐ I sometimes share some interesting facts ☐ I rarely talk to them about it ☐ My parents aren't interested

20. Has your parents' behavior changed since you joined the eco-club?

☐ Yes, significantly ☐ Yes, a bit ☐ No ☐ Not sure

21. What do you like the most about the eco-club? (choose up to three options)

☐ Learning about new technologies ☐ Applied projects ☐ Spending time with friends ☐ The opportunity to affect the environment of the place I live in ☐ Role-playing and interactive classes ☐ Preparation for choosing a career/educational program after school ☐ Gaining new skills

Appendix 2

Parent survey

Instructions: Dear parents! We invite you to take part in our study on the influence of eco-clubs on environmental education. Your answers will help improve ecological education in schools.

Block 1. Demographic information

1. Age: _____

2. Education: _____

☐ Secondary ☐ Secondary vocational ☐ Higher ☐ Academic degree

3. How many children in your family participate in the eco-club? _____

Block 2. Environmental knowledge

4. Where do you get information about environmental issues? (select all that apply)

☐ My child ☐ Mass media ☐ Internet ☐ Work ☐ Friends ☐ Other

5. Have you learned anything new about ecology from your child in the past 6 months?

☐ Yes, a lot ☐ Yes, a bit ☐ No ☐ Not sure

Block 3. Changes in family practices

6. What environmental practices has your family started using since your child joined the eco-club?

7. What do you think should change in the behavior of people in your region to improve the environmental situation?

Appendix 3

Expert assessment form

Expert: _____ **Organization:** _____ **Date:** _____

Instructions: Please rate the importance of the following factors for the effectiveness of eco-clubs on a 5-point scale (1—completely unimportant, 5—extremely important).

No.	Factor	Score (1–5)
1	Support from the school administration	
2	Qualification of the eco-club coordinator	
3	Material and technical equipment	
4	Connection with the local community	
5	Parent engagement	
6	Practical focus of activities	
7	Application of modern technologies	
8	Inter-school cooperation	
9	Support from local authorities	
10	Integration with the educational program	

Appendix 4

Student focus group guide

Introduction (5 minutes)

- Introduction of the moderator
- Explanation of the goals and rules of the discussion
- Obtaining consent for audio recording

Primary questions (60–70 minutes)

1. Experience of participating in the eco-club (15 minutes)

- What first brought you to the eco-club?
- What events at the eco-club were the most memorable to you?
- What has changed in your understanding of environmental problems?

2. Impact on the family (20 minutes)

- Do you talk about the activities of the eco-club at home?
- How do your parents respond to your environmental initiatives?
- Please give some examples of how your parents' behavior changes under your influence.

3. Barriers and opportunities (15 minutes)

- What stops you from participating in environmental initiatives more actively?
- What support would you like to receive from the school or your parents?

4. Vision of the future (10 minutes)

- How do you see the further development of eco-clubs?
- What could be improved in their work?

Conclusion (5 minutes)

- Thanking the participants
- Information on the further use of the results